

Relational Roles and Evolution: Role-Scoped Evolution Governance, Evolution-Triggered Role Changes, and Vertical Evolution Across Level Role Scopes

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

This note is the eighty-seventh in Phase B2 of Series B. It is the fourth of five notes decomposing B1.17 (relational roles), following B2.84 (relational roles integrating frame), B2.85 (purpose-defined level determination), and B2.86 (same entity at multiple levels). B2.88 (relational roles verification) will close the B1.17 decomposition thread. It does not introduce new commitments; its sole contribution is to formalize, in operational form, how relational roles per B1.17 interact with the three evolution mechanisms per B1.13–B1.15 and with bidirectional evolution per B1.16.

Abstract

Paper 2 establishes relational roles (B1.17) and three evolution mechanisms in productive tension (B1.13–B1.15) as independent architectural commitments. This note formalizes their interaction as the operational specification of role-scoped evolution governance. Five interaction patterns compose the formalization: role-scoped evolution, in which each mechanism operates at the level scope corresponding to the entity’s role per B2.85; evolution-triggered role changes, in which accumulated evolution at one scope may warrant governed redetermination at a different level; birth triggered by evolution-role interaction, in which vertical evolution creating new aspects or Selves initiates birth per B1.09 with level determination as a birth specification requirement per B2.40; vertical evolution connecting level scopes per B1.16, in which the relational character of roles means cross-scope influence is propagation between role-based scopes rather than between fixed layer boundaries; and multi-role evolution independence per B2.86, in which an entity occupying multiple level roles evolves independently at each scope, with vertical evolution per B1.16 as the governed pathway for deliberate cross-scope influence. The note articulates what makes role-scoped evolution architecturally distinctive, states its biological analog as conceptual scaffold, identifies the inherited

Paper 1 commitments, states operational implications, and names its limits.

1. Why relational roles and evolution requires standalone formalization

B1.17 establishes that structural roles in a CKS Self are relational and purpose-defined rather than intrinsic — the same underlying CKS artifact participates as a cell in one arrangement, as part of an aspect in another, or as a component of different aspects simultaneously, depending on what the structure is for. B1.13–B1.15 establish three evolution mechanisms operating in productive tension: instinct evolution (undirected, mutation-like), DNA evolution (directed selection on stabilized orchestration content), and action-feedback evolution (closing the loop from lived experience back into DNA). B1.16 establishes that evolution operates on two axes — horizontal (content within existing structure) and vertical (structural reorganization itself) — with vertical evolution making structural composition directly available as a governed operation.

Each of these commitments stands independently and has been formalized in prior Phase B2 notes. What has not yet been formalized as a standalone architectural derivation is how they interact: when an entity occupies a relational role at a particular level scope, how does each evolution mechanism apply? What scope does each mechanism act on? Can evolution cause an entity to change its role? How does vertical evolution per B1.16 connect role-based scopes? How does an entity occupying multiple roles simultaneously manage independent evolution governance at each scope?

This note formalizes that interaction as a coherent specification. Its strategic purpose in the prior-art chain is to establish, as a dated public record, the operational territory covering role-scoped evolution governance, evolution-triggered role change under governed redetermination, vertical evolution as cross-scope propagation between role-based scopes, and multi-role evolution independence with vertical evolution as the deliberate cross-scope pathway. The note occupies the eighty-seventh position in Phase B2, functioning within the wave structure as the operational specification that closes the substantive B1.17 decomposition before B2.88's verification treatment.

2. The architectural interaction precisely stated

Role-scoped evolution. Each of the three evolution mechanisms operates at the level scope corresponding to the entity's role as determined per B2.85. Scope is not fixed by physical location in a stack; it is determined by what purpose the entity serves in the structural arrangement.

For instinct evolution, the mutation-like mechanism operates at cell scope on the instinct and infrastructure layer of cell-level entities. An aspect-level entity that coordinates cells also experiences instinct evolution effects, but those effects arrive through the cells the aspect comprises; the mutation governance operates at cell scope per the cells' roles, not at aspect scope per the coordinating entity's role. For DNA evolution, directed selection per B1.14 applies to DNA scope at whatever level scope the entity's role determines: cell-role DNA scope when the entity acts as a cell, aspect-role DNA scope

when the entity coordinates cells for a purpose, Self-role DNA scope when the entity integrates aspects as a unified whole. For action-feedback evolution, the mechanism examines action-layer operational patterns at the level scope corresponding to the entity's role: cell-level action-feedback surfaces from the entity's task execution patterns at cell scope; aspect-level action-feedback surfaces from the entity's coordination patterns across cells; Self-level action-feedback surfaces from the entity's integration patterns across aspects.

Evolution-triggered role changes. Role-scoped evolution creates the operational condition under which evolution may surface a need for role redetermination. Two primary evolution-triggered paths arise directly from the source paper's commitments.

Cell-to-aspect evolution-triggered: a cell that accumulates coordination responsibilities through operational evolution — through DNA evolution adding coordination capabilities, through action-feedback surfacing patterns that resemble aspect-level integration, or through directed selection building out multi-cell governance content — may warrant redetermination as an aspect. The accumulated evolution does not automatically produce the role change; it produces the condition that may warrant governed redetermination per B2.85. The redetermination requires a governance decision with the new level determination authored per A2.04, with the prior level determination preserved per A6.02's retroactivity commitment.

Aspect-to-Self evolution-triggered: an aspect that accumulates integration responsibilities across multiple subordinate aspects may similarly warrant redetermination as a Self. The same governed redetermination process per B2.85 applies. The evolution generates the evidential basis; governance decides.

Birth triggered by evolution-role interaction. When vertical evolution per B1.16 creates new aspects or Selves through structural reorganization — when cells are assembled into a newly governed aspect, or when aspects are assembled into a newly governed Self — birth per B1.09 is triggered for the new entities. These births include level determination per B2.85 as part of birth specification requirements per B2.40. The birth event is the governed origination of the new entity at its level scope; the role determination is part of what birth specifies.

Vertical evolution connecting level scopes. Vertical evolution per B1.16 operates between level scopes. The relational character of roles per B1.17 means this is precisely propagation between role-based scopes, not propagation between fixed layer boundaries. Upward propagation carries cell-scope evolution outcomes into aspect-scope consideration: what cells have learned or accumulated through DNA or action-feedback evolution becomes available, through governance, to inform aspect-scope evolution decisions. The structural arrangement that vertical evolution can reorganize is itself substrate content — orchestration substrates that define composition — and is human-governed in the same way DNA content is governed. Vertical evolution does not bypass governance; it is a governed substrate-edit operation on composition-defining content.

Multi-role evolution independence. For entities occupying multiple level roles simultaneously per B2.86, evolution at each level scope is independent by default. Directed selection at cell scope does not automatically propagate changes to the entity's aspect-scope DNA. Action-feedback at cell scope does not automatically produce aspect-scope proposals. The independence is the default; vertical evolution per B1.16 is the deliberate, governed pathway through which evolution at one scope may be carried into another scope through human governance decisions.

Role-consistent governance. Evolution governance at each level scope is consistent with the entity's role at that scope. Cell-level evolution uses the governance framework applicable to cell-level entities; aspect-level evolution uses aspect governance; Self-level evolution uses Self governance. The role determines the applicable governance framework, not the physical artifact type.

3. What makes role-scoped evolution architecturally distinctive

Conventional AI architectures do not have role-scoped evolution. When an AI component evolves — through fine-tuning, retraining, or capability update — the evolution applies to the component as a whole, without differentiation by the component's role in a larger coordination arrangement. The component is what it is; its role in a workflow is external metadata, not an architectural determinant of how evolution applies.

CKS role-scoped evolution means the same substrate artifact can evolve differently at different level scopes, with evolution governance calibrated to each role. An artifact participating as a cell in one arrangement and as part of an aspect in another has distinct evolution governance at each scope. Its cell-scope DNA can evolve under cell governance while its participation in the aspect remains governed under aspect governance. The two evolutions are independent by default and connected only through the governed vertical evolution pathway.

The evolution-triggered role change is particularly distinctive. In conventional AI architectures, component roles are fixed at design time or assigned by workflow configuration; they do not change in response to operational evolution. In CKS, evolution can cause an entity to accumulate the characteristics that warrant a different level determination — and that warrant change is then the subject of governed redetermination per B2.85. Evolution does not determine the role; governance determines the role, informed by what evolution has produced. This distinction — evolution as evidence, governance as decision — is what makes the role change both architecturally responsive and architecturally controlled.

4. The biological analog as conceptual scaffold

The relational-roles-and-evolution interaction is structurally analogous to biological developmental cell fate changes driven by evolution. In biology, cells can change their functional roles under evolutionary pressure — differentiated cells that accumulate regulatory capacities through evolutionary time may come to occupy new organizational roles in more complex organisms. The process operates on the

cells' accumulated capabilities; which organizational role those capabilities warrant is determined by the developmental program that governs organizational structure.

The analogy is also structurally analogous to epigenetic evolution across organizational levels, where cells at different levels of biological organization accumulate changes at their respective organizational scopes and those changes remain loosely coupled across levels. What biology lacks is precise governance of the cross-level propagation: biological cross-level evolution propagation operates through natural selection acting on fitness outcomes, not through explicit governed decisions over what propagates and what does not.

CKS's governed vertical evolution per B1.16 is the architectural property that exceeds the biological analog: cross-scope evolution influence is a governed substrate-edit operation, explicitly decided upon and authored by humans with the appropriate authority scope. The evolution-triggered role change exceeds the analog in the same direction: the governance decision per B2.85 is explicit, authored per A2.04, and preserving the prior determination per A6.02, rather than implicit in competitive selection. The analog functions as conceptual scaffold. The architectural substance is role-scoped governed evolution with role change possible through evolution-triggered governed redetermination.

5. Inherited Paper 1 commitments

A1.01 (human-governed). Evolution-triggered role changes are governed: the governance decision is required; the role change does not occur automatically based on observed evolution alone. Evolution provides the evidential basis; human authority per A1.01's three-rights definition determines the outcome.

B1.20 (recursive Paper 1 commitments). Paper 1's commitments apply at each level scope independently. Role-scoped evolution governance inherits this recursion: cell-scope evolution governance instantiates the full Paper 1 commitment set at cell scope; aspect-scope evolution governance instantiates it at aspect scope; Self-scope evolution governance instantiates it at Self scope. The recursive application is what makes role-scoped governance architecturally coherent across all three levels rather than ad hoc.

A6.02 (retroactivity). Prior role determinations are preserved when evolution-triggered role changes occur. When an entity is redetermined from cell to aspect, the cell-level determination history — governance decisions, authored rules, evolution records — remains preserved as prior-determination substrate content. The new determination starts a new governance scope; it does not overwrite or retroactively recharacterize the prior scope.

B1.16 (bidirectional evolution). Vertical evolution is the governed mechanism that connects role-based scopes. The relational character of roles means vertical evolution is propagation between role-based scopes, not propagation between fixed layer boundaries. Horizontal evolution operates within existing role-based scope; vertical evolution reorganizes the composition that determines which

scopes exist and how they relate.

A2.04 (rule authoring). Level determinations produced through evolution-triggered redetermination are authored: the new level determination is a governed substrate edit, authored by humans with the authority scope appropriate to the redetermination. The authoring requirement applies both to the new determination and to the governance decision that precedes it.

A2.40 (provenance). Evolution-triggered role changes are recorded as substrate content with provenance metadata. The decision to redetermine, the evidential basis from accumulated evolution, the prior determination reference, and the new determination specification are all substrate content subject to the six-field provenance requirement.

6. Operational implications

Governance processes should monitor evolution at each scope for indicators that may warrant role change consideration. The operational question is: at what accumulated evolution threshold does governance review become appropriate? The formalization does not specify a threshold — that is a deployment governance decision — but it specifies that governance review must occur before any role change can proceed. Deployments that accumulate cells developing coordination capabilities through operational DNA evolution and action-feedback accumulation may, over their operational lifecycle, encounter conditions where cell-to-aspect redetermination becomes a governance question.

Multi-role entities per B2.86 require evolution governance infrastructure at each scope they occupy simultaneously. The independence of evolution at each scope is not purely administrative; it is architectural. Governance processes that conflate cell-scope evolution governance with aspect-scope evolution governance for the same artifact introduce the scope confusion that role-scoped governance is designed to prevent.

Vertical evolution per B1.16 is the deliberate cross-scope pathway, not an implicit coupling. Deployments that wish to carry cell-scope evolution outcomes into aspect-scope governance decisions must do so through the governed vertical evolution process: explicit substrate edits to composition-defining content, authored by humans with the appropriate authority scope, subject to provenance requirements per A2.40.

When evolution-triggered role change proceeds through governed redetermination, the birth per B1.09 triggered for newly created entities must include level determination per B2.85 as a birth specification requirement per B2.40. Deployments that have established birth specification processes should verify that those processes accommodate the role determination step as a required specification component, not an optional downstream addition.

Mature deployments — those with operational lifetimes long enough for substantial DNA evolution and action-feedback accumulation — are the most likely to encounter evolution-triggered role change conditions. The formalization positions governance review of accumulated evolution as a regular

governance activity rather than an exceptional intervention.

7. Limits

Role-scoped evolution does not mean isolated scopes. The independence of evolution at each scope is independence of governance, not isolation of operational effect. An entity participating in multiple arrangements experiences the effects of evolution at each scope in its operational behavior. Vertical evolution per B1.16 is the governed mechanism for deliberate cross-scope influence, but it is not the only pathway through which evolution at one scope affects operational outcomes at another.

Evolution-triggered role change is not automatic. The formalization does not establish any threshold at which accumulated evolution automatically produces a role change. Evolution provides the condition that may warrant redetermination; governance decides. A cell that has accumulated substantial coordination capabilities through DNA evolution remains a cell in its current arrangement until governed redetermination per B2.85 occurs.

Prior determination is preserved, but new governance scope starts fresh. A6.02's retroactivity commitment preserves the prior level determination and its associated governance history. The new determination, however, starts a new governance scope. Prior cell-scope DNA does not automatically become aspect-scope DNA; prior cell-scope orchestration rules do not automatically govern the new aspect scope. The new scope is built through the authored governance process per A2.04.

Role-scoped evolution does not introduce new mechanisms. The same three mechanisms per B1.13–B1.15 apply at every level scope. Role-scoped evolution governance means the same mechanisms operate with governance calibrated to the entity's role at each scope; it does not mean different mechanisms for different roles.

Multi-role evolution independence is the default, not an absolute constraint. Vertical evolution per B1.16 is the deliberate pathway for cross-scope influence when governance determines that carrying evolution across scopes is warranted. The default independence can be bridged through this governed pathway; it cannot be bridged through informal operational coupling that bypasses governance.

8. Architectural interaction test

A deployment instantiates the relational-roles-and-evolution interaction per this note if and only if: each evolution mechanism operates at the level scope corresponding to the entity's role as determined per B2.85, not at the scope of the physical artifact type; evolution-triggered role changes proceed only through governed redetermination per B2.85, with governance decision required and prior determination preserved per A6.02; vertical evolution per B1.16 is the governed pathway for cross-scope evolution influence; multi-role entities per B2.86 have independent evolution governance at each occupied scope; and births triggered by evolution-role interaction include level determination as a birth specification requirement per B2.40.

9. Position in the B1.17 decomposition and Phase B2 progression

This note is the fourth of five notes decomposing B1.17. B2.84 established the integrating frame for relational roles — that roles are relational and purpose-defined, that the same artifact can participate at multiple levels, and that B1.17 is the parent commitment for the decomposition thread. B2.85 formalized purpose-defined level determination — how governance determines which level scope an entity occupies and what the governed determination process looks like. B2.86 formalized the same-entity-at-multiple-levels case — the operational specification of what it means for one artifact to simultaneously occupy multiple level roles and what governance obligations that creates. B2.87 (this note) formalizes how relational roles interact with the three evolution mechanisms, providing the operational specification that makes role-scoped evolution governance a named architectural property with formalized limits, triggers, and inheritance chain.

B2.88 will close the B1.17 decomposition thread with relational roles verification — the operational specification of how verification substrates and governance processes confirm that relational role assignments remain warranted as evolution and operational conditions change. Following B2.88, Phase B2 notes continue with the B1.18 content-domain decomposition beginning at B2.89.

The naming of this note as a standalone derivation — rather than subsuming the interaction under either the evolution mechanism notes (B2.67–B2.83) or the relational roles frame (B2.84) — establishes role-scoped governed evolution, evolution-triggered role change, and vertical evolution as cross-scope role propagation as distinct patentable territory in the prior-art chain. Each of these architectural properties is derivable from Paper 2’s commitments; none is named as a standalone operational derivation in the source paper; their formalization here as dated public prior art narrows the territory within which any party could claim novel invention in this space.

Source

Li, Wenxin. “The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.” April 2026.

Li, Wenxin. “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.” April 2026.

Self-Citation

This note is part of the CKS derivation note series. Related notes in the B1.17 decomposition thread: B2.84 (relational roles integrating frame), B2.85 (purpose-defined level determination), B2.86 (same entity at multiple levels), B2.88 (relational roles verification, forthcoming). Directly load-bearing evolution notes: B1.13–B1.15 (three evolution mechanisms), B1.16 (bidirectional evolution). Directly

load-bearing governance notes: A1.01 (human-governed definition), B1.20 (recursive Paper 1 commitments), A6.02 (retroactivity), A2.04 (rule authoring), A2.40 (provenance).